

Bun Test Runner

Source: <https://bun.sh/docs/test>

Bun ships with a fast, built-in, Jest-compatible test runner.

Features

- TypeScript and JSX support
- Lifecycle hooks
- Snapshot testing
- UI & DOM testing
- Watch mode with `--watch`
- Script pre-loading with `--preload`

Run Tests

```
bun test
```

Writing Tests

```
import { expect, test } from "bun:test";

test("2 + 2", () => {
  expect(2 + 2).toBe(4);
});
```

Test files match patterns:

- `*.test.{js|jsx|ts|tsx}`
- `*_test.{js|jsx|ts|tsx}`
- `*.spec.{js|jsx|ts|tsx}`
- `*_spec.{js|jsx|ts|tsx}`

Filtering Tests

```
bun test <filter> # Filter by file path
bun test --test-name-pattern addition # Filter by test name
bun test ./test/specific-file.test.ts # Run specific file
```

Common Options

```
bun test --timeout 20 # Per-test timeout in ms (default: 5000)
bun test --bail # Stop after first failure
bun test --watch # Watch mode
bun test --rerun-each 100 # Run each test N times
bun test --coverage # Generate coverage
```

Lifecycle Hooks

```
import { beforeAll, beforeEach, afterEach, afterAll } from "bun:test";

beforeAll(() => { /* runs once before all tests */ });
beforeEach(() => { /* runs before each test */ });
afterEach(() => { /* runs after each test */ });
afterAll(() => { /* runs once after all tests */ });
```

Mocks

```
import { test, expect, mock } from "bun:test";

const random = mock(() => Math.random());

test("random", () => {
  const val = random();
  expect(random).toHaveBeenCalled();
  expect(random).toHaveBeenCalledTimes(1);
});
```

Snapshots

```
import { test, expect } from "bun:test";

test("snapshot", () => {
  expect({ a: 1 }).toMatchSnapshot();
});
```

Update snapshots:

```
bun test --update-snapshots
```

CI/CD Integration

GitHub Actions:

```
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: oven-sh/setup-bun@v2
      - run: bun test
```

JUnit XML reports:

```
bun test --reporter=junit --reporter-outfile=./bun.xml
```